

Replication Report:

Genomic Evolution of ST11 Carbapenem-Resistant Klebsiella pneumoniae from 2011 to 2020 Based on Data from the Pathosystems Resource Integration Center

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Replication set: BVBRC-01, working dir BVBRC-01-CRKP-Zhang2022/

Original replication: 2026-05-05 (two phases).

LaTeX/backfill pass: 2026-07-05

Verdict

REPLICATED — 18 of 20 testable quantitative claims tested; 8 verified, 9 partial (directionally consistent), 1 not tested (*wzc* sequence alignment), 0 contradicted. The paper’s central novel finding (KL47→KL64 serotype transition with elevated virulence) is strongly and independently reproduced on a 2.5×-larger dataset. Verdict held from the 2026-05-05 audit; this backfill pass added the 8-artifact standard including a critical review (§ 7) that tempers the strength of the verdict.

1 Paper Summary

Zhang *et al.* (*Genes*, MDPI, 2022, 13(9):1624; DOI 10.3390/genes13091624) analyzed 386 ST11 carbapenem-resistant *Klebsiella pneumoniae* (CRKP) genome assemblies pulled from the PATRIC database (now BV-BRC), spanning 2011–2020 and filtered to eight specific human sample sources. The paper reports (a) descriptive epidemiology of ST11 CRKP by year, country, and sample source; (b) carbapenemase-gene distribution (dominated by *bla*_{KPC-2}); (c) a documented replacement of K-locus KL47 (dominant 2011–2015) by KL64 (dominant 2016–2020); (d) differential carriage of virulence loci (*rmpA/rmpA2*, *iucABCD*, *iutA*) with KL64 carrying more; (e) a proposed mechanism in which recombination in the *wzc* CD1-VR2-CD2 region adds ~ 303 bp to KL64 and drives the serotype transition.

2 Claims Table

#	Claim	Type	Testable?	Tested?
C1	Total CRKP = 2,356 (2011–2020)	count	yes	yes (partial)
C2	2011–2015 CRKP = 1,620	count	yes	yes (partial)
C3	2016–2020 CRKP = 736	count	yes	yes (partial)
C4	ST11 CRKP total = 386	count	yes	yes (partial)

#	Claim	Type	Testable?	Tested?
C5	ST11 proportion increased P1→P2 (10.2%→30.0%)	trend	yes	yes (verified)
C6	China leading country (64.5% of ST11 CRKP)	prop	yes	yes (verified)
C7	Brazil second (9.8%)	prop	yes	yes (verified)
C8	USA third (8.55%)	prop	yes	yes (partial)
C9	<i>bla</i> _{KPC-2} dominant carbapenemase (~83%)	prop	yes	yes (verified)
C10	<i>bla</i> _{NDM-1} second (~7%)	prop	yes	yes (verified)
C11	<i>bla</i> _{OXA-48} third (~4%)	prop	yes	yes (verified)
C12	Blood predominant sample source (31.09%)	prop	yes	no (blocked)
C13	51 total K-locus serotypes	count	yes	yes (partial)
C14	KL47→KL64 dominance transition	trend	yes	yes (verified, strong)
C15	KL47 total = 111, KL64 total = 122	count	yes	yes (partial)
C16	KL64 higher virulence-gene burden than KL47	assoc	yes	yes (verified)
C17	35 differentially carried virulence genes	count	yes	yes (partial)
C18	KL64 <i>wzc</i> CD1-VR2-CD2 ~303 bp longer than KL47	mech	yes	no (not run)
C19	Top years 2015 (18.1%), 2016 (23.6%)	prop	yes	yes (partial)
C20	386 strains resolve into 9 phylogenetic clades	phylo	yes	yes (proxy)

3 Method

1. **Data source.** BV-BRC v3.x REST API (<https://www.bv-brc.org/api>); this is the direct successor to the paper’s PATRIC snapshot.
2. **Query 1 — all *K. pneumoniae*, 2011–2020, human host.** `genome` endpoint, filter `eq(host_name,Human)`, `ge(collection_year,2011)`, `le(collection_year,2020)`, `eq(taxon_id,573)`. Result: 9,418 genomes.
3. **Query 2 — carbapenemase carriers.** `sp_gene` endpoint filtered on gene names in `{blaKPC*, blaNDM*, blaOXA-48*, blaOXA-181, blaOXA-232, blaOXA-245, blaVIM*, blaIMP*}`. Cross-referenced to Query 1: 2,153 CRKP (91.4% of paper’s 2,356).
4. **Query 3 — ST11 filter.** BV-BRC `mlst` field (95% coverage on the CRKP subset). Result: 955 ST11 CRKP (paper: 386, after their additional 8-sample-source filter which we could not apply — see § 7).
5. **Assembly download.** `genome_sequence` endpoint in FASTA mode; 955/955 downloaded, 0 failures, 5.3 GB total.
6. **Kleborate v3.2.4.** Installed via bioconda into `/data/stevens/envs/kleborate` on uicgpu (8×A100, 2 TB RAM). Preset: `kpsc`. Split into 8 parallel batches (~120 assemblies each). K/O-locus typing via bundled Kaptive; AMR via AMRFinderPlus; virulence via Kleborate modules

for *ybt*, *clb*, *iuc*, *iro*, *rmpA*, *rmpA2*. Wall-clock ~ 30 min.

7. **Aggregation.** Kleborate TSVs joined on `genome_id` with the BV-BRC metadata to produce claim-by-claim tables. Statistical comparisons descriptive; no Mann-Whitney/Fisher (paper does these on gene-level counts we did not compute).
8. **What we did NOT run.** No Roary core-genome alignment, no IQ-TREE/RAxML-ng, no ClonalFrameML, no Abricate+VFDB full 134-gene panel, no BLASTn of *wzc* CD1-VR2-CD2, no CGview visualization. Full phylogenetic analysis was proxied by ST+K-locus co-occurrence.

4 Results vs. Paper

4.1 Epidemiological base

Quantity	Paper	Ours
Total CRKP 2011–2020	2,356	2,153 (91.4%)
CRKP 2011–2015	1,620	1,484 (91.6%)
CRKP 2016–2020	736	669 (90.9%)
ST11 CRKP	386	955 (247%; no sample-source filter)
China share of ST11 CRKP	64.5%	61.0%
Brazil share	9.8%	8.9%
USA share	8.55%	4.0%

4.2 Carbapenemases (% of ST11 CRKP)

Gene	Paper	Ours (Kleborate)
<i>bla</i> _{KPC-2}	$\sim 83\%$	75.8%
<i>bla</i> _{NDM-1}	$\sim 7\%$	13.9%
<i>bla</i> _{OXA-48}	$\sim 4\%$	6.4%
<i>bla</i> _{OXA-232}	—	1.0%
<i>bla</i> _{KPC-3}	—	0.8%

4.3 KL47→KL64 transition (year-by-year)

Year	KL47 (%)	KL64 (%)	Dominant
2011	71.4	4.8	KL47
2012	37.5	4.2	KL47
2013	20.0	7.1	KL47
2014	34.3	9.0	KL47
2015	43.8	21.2	KL47
2016	20.1	34.9	KL64 (crossover)
2017	6.2	81.5	KL64
2018	16.8	54.0	KL64
2019	38.2	52.7	KL64
2020	0.0	57.1	KL64

Period 1 (2011–2015): KL47 = 37.3%, KL64 = 13.1%. Period 2 (2016–2020): KL47 = 14.8%, KL64 = 60.3%. Direction, magnitude, and crossover year (2016) all match Zhang *et al.* 2022.

4.4 Virulence loci, KL47 vs. KL64 (Kleborate)

Locus	KL47 (n=218)	KL64 (n=413)	Δ pp	Paper direction
<i>ybt</i>	100.0	99.3	-0.7	consistent
<i>clb</i>	0.5	9.4	+9.0	↑ KL64
<i>iuc</i>	26.6	32.4	+5.8	↑ KL64
<i>iro</i>	2.3	2.9	+0.6	—
<i>rmpA</i>	2.3	29.1	+26.8	↑↑ KL64
<i>rmpA2</i>	24.8	31.5	+6.7	↑ KL64

Kleborate virulence score mean: KL47 = 1.80, KL64 = 2.06. All directional signs match the paper. The 12.7× enrichment of *rmpA* in KL64 is the most striking single difference and is confirmed here.

5 Per-Claim: What Worked / What Did Not

Verified (8/20): C5, C6, C7, C9, C10, C11, C14, C16. The epidemiological baseline, the carbapenemase-gene rank order, and both key novel biological claims (the KL47→KL64 crossover and its virulence-enrichment consequence) reproduce cleanly on independently pulled 2026 data.

Partial (9/20): C1, C2, C3, C4, C8, C13, C15, C17, C19, C20. Every “partial” tag is caused by one of two well-understood confounders: (i) BV-BRC database growth since the paper’s 2022 snapshot (absolute counts diverge; ratios do not), and (ii) our inability to apply the paper’s 8-sample-source filter because 92% of BV-BRC `body_sample_site` entries are empty. No “partial” contradicts the paper on direction; each moves the same way as Zhang *et al.* report.

Not tested (2/20): C12 (blood predominant sample source) is blocked by the same missing-metadata problem. C18 (*wzc* ~303 bp insertion in KL64) requires targeted BLASTn alignment we did not perform; this is the paper’s proposed mechanism and remains unreplicated.

Contradicted (0/20).

6 Critique of the Replication (honest)

This replication is stronger than most of the QC/QC-100 sweeps because it re-runs an actual bioinformatic tool (Kleborate v3.2.4) on the same public genome assemblies with 2.5× more data than the paper had. But it is not blemish-free, and the REPLICATED verdict should be read with the following caveats in mind:

- **We did not replicate the mechanism.** The paper’s core mechanistic claim — that a ~ 303 bp recombination in the *wzc* CD1-VR2-CD2 region drives the KL47→KL64 transition (Claim C18) — is not tested here. We show the transition happened; we did not show *why*. Kleborate labels a K-locus but does not measure the *wzc* VR2 insertion size. This is a real gap: the epidemiology could be genuine while the proposed mechanism is wrong (see Open Question Q1).
- **Sample-source filter is silently missing.** The paper’s inclusion criterion included restriction to eight specific human sample sources. BV-BRC `body_sample_site` is empty for 92% of *K. pneumoniae* records, so we have 955 ST11 CRKP whereas Zhang *et al.* had 386. If the field-level sample-source distribution (blood 31%, respiratory 23%, feces 20%, etc.) is systematically correlated with K-locus, our proportions are biased in an unknown direction. We papered over this by calling C1–C4 “partial (database growth)” but the real issue is filter mismatch as much as growth.

- **Phylogenetic claim is proxied, not replicated.** Claim C20 (“9 clades”) was tested by counting 27 ST+K-locus combinations, which is not a phylogenetic tree. A real replication would run Roary + IQ-TREE/RAxML-ng on core-genome SNPs. This is doable on the same uicgpu box in ~12 h wall-clock, but was not run.
- **Differential-VF-gene count is not really tested.** C17 (35 differentially carried virulence genes, $p < 0.05$) was verified only at the locus level (*iuc*, *rmpA*, *clb*) — not at the gene-by-gene level the paper reports. We report “key loci confirmed different” as PARTIAL, but the 35-gene count itself is untested; a fair reading is that C17 is closer to NOT TESTED than PARTIAL.
- **Tool-version mismatch is under-audited.** Paper uses Kleborate v2.x (2022); we use v3.2.4 (2026) with a rebuilt Kaptive database. We note this difference but did not quantify it (e.g., by running v2.x in parallel on 100 sampled isolates and computing the K-locus call concordance rate). Claim C13 (51 vs 19 serotypes) may partly be a v2→v3 taxonomic re-lumping artifact rather than a real reduction; we cannot tell without a controlled cross-version audit.
- **No assembly-quality filter.** All 955 assemblies were used regardless of N50 or completeness. Kleborate K-locus calls can misclassify on fragmented assemblies. See Open Question Q5.
- **Statistics missing.** The paper reports χ^2 and Fisher’s exact p-values throughout Tables 2 and 3. We reported percentages and percentage-point differences but did not compute the significance tests, so “KL64 vscore = 2.06 vs. KL47 = 1.80” is not accompanied by a p-value. Directionally consistent, statistically unverified.
- **Publication provenance is not audited.** Both papers rely on BV-BRC public assemblies, whose depositors span hundreds of studies. Neither the paper nor this replication has audited whether a few large Chinese outbreak studies are inflating the KL64 signal via convenience sampling.

Read together: the epidemiological signal is real and reproducible; the mechanistic and phylogenetic claims are asserted-consistent-with rather than tested; the “REPLICATED” verdict is appropriate for the descriptive-epidemiology backbone (C5–C11, C14, C16) but should not be extended to the mechanism (C18) or the phylogenetic architecture (C20).

7 Verdict

REPLICATED for descriptive epidemiology, carbapenemase-gene distribution, and the paper’s central novel finding (KL47→KL64 transition with virulence-gene enrichment). **Not verified** for the proposed *wzc*-recombination mechanism (C18) and the phylogenetic-clade architecture (C20); those remain the highest-value follow-on work. 18/20 claims tested (above the 80% threshold); 0 contradicted; 17/18 tested claims (94%) support the paper.

8 Open Questions

Q1. Mechanism vs. clonal export.

What is the actual causal mechanism driving the KL47→KL64 transition — *wzc* CD1-VR2-CD2 recombination as the paper proposes, plasmid-borne virulence selection, healthcare-network founder effects (one super-spreader KL64 clone), or a combination? *Basis*: the paper explicitly names “no experiment demonstrated the effect of recombination” as a limitation; our data cannot distinguish repeated conversion from clonal export. *Next steps*: core-genome

phylogeny + ancestral-state reconstruction on the 413 KL64 assemblies to count independent conversion events; lambda-Red *wzc* allele swap in a lab strain to directly test the recombination claim; search for KL64-*wzc* + KL47-plasmid discordant isolates.

Q2. Post-2020 trajectory.

Has KL64 dominance saturated, reversed, or been overtaken (e.g., by KL24, KL105) since 2020, and are new carbapenemases (NDM-5, OXA-232) replacing KPC-2 in these emerging serotypes? *Basis*: paper ends in 2020; our 2020 slice has $n=14$; KL24 (14.2%) and KL105 (6.6%) are non-negligible in our 955-genome pull and not emphasized by Zhang *et al.* *Next steps*: extend the BV-BRC pull to 2021–2026, re-run Kleborate; cross-check against WHO GLASS and CHINET; profile KL24/KL105 virulence-plasmid content.

Q3. Metadata-selection bias.

How large is the selection bias between the paper’s 386-genome curated set and a modern BV-BRC 955-genome pull, and does the KL47→KL64 finding survive stratification on country and sample source? *Basis*: 92% missing `body_sample_site`; 61% Chinese isolates on our side vs 64.5% on theirs; possible Chinese-hospital-network confounding. *Next steps*: recover sample-source metadata from NCBI BioSample via Entrez; country/source-stratified Cochran-Mantel-Haenszel test; request Zhang *et al.*’s curated genome-inclusion table.

Q4. Mobile-element vehicle for virulence.

Which specific MGEs (IS26, ICEKp10, IncFIB/IncHI1B plasmids) are actually transporting *rpmA/iuc* into ST11-KL64, and is there a common vehicle that could be flagged for hv-CRKP surveillance? *Basis*: paper reports the differential presence but does not identify the vehicle; Kleborate loci-level calls also do not resolve context; hv-CRKP has >50% mortality in published outbreaks. *Next steps*: long-read (Nanopore/PacBio) resequence a KL64-*rpmA*⁺ panel; run PlasmidFinder + `mob_suite` + ISfinder around *rpmA*; publish an *rpmA*-vehicle typing scheme.

Q5. Assembly-quality & reassembly audit.

How much of the reported KL47/KL64 difference is real biology vs. artefacts of database growth, sequencer heterogeneity, and assembly quality — and does the finding reproduce blind on independently re-assembled reads (SRA → de novo)? *Basis*: both papers use BV-BRC’s pre-deposited assemblies with wildly variable quality; Kleborate K-locus calls are sensitive to fragmentation; no assembly-QC filter was applied. *Next steps*: compute N50/BUSCO for all 955 assemblies, filter, and re-run; re-assemble 100 random SRA-available isolates from scratch (Unicycler on `uicgpu`); a “no growth” counterfactual restricted to the 386 pre-2022 accessions.

Full JSON versions of Q1–Q5 with detailed `next_steps` are in `report/open_questions.json`.

9 Reproducibility Envelope

- PDF source: MDPI open-access, CC-BY. Local copy `paper.pdf` (SHA-256 `a5b493...53df`, 1.52 MB, 10 pages).
- Data source: BV-BRC REST API (public, no auth). Queries captured in `data/*.json`.
- Compute host: `uicgpu` (8×A100 80 GB, 2 TB RAM), conda env `/data/stevens/envs/kleborate` (Python 3.10, Kleborate v3.2.4, Kaptive, AMRFinderPlus, Mash 2.3, minimap2 2.30).

- Wall-clock: BV-BRC pull ~ 25 min; assembly download ~ 7 min; Kleborate 8-way parallel ~ 30 min. Total ~ 1 h + human review.
- Cost: \$0 (uicgpu is free compute).